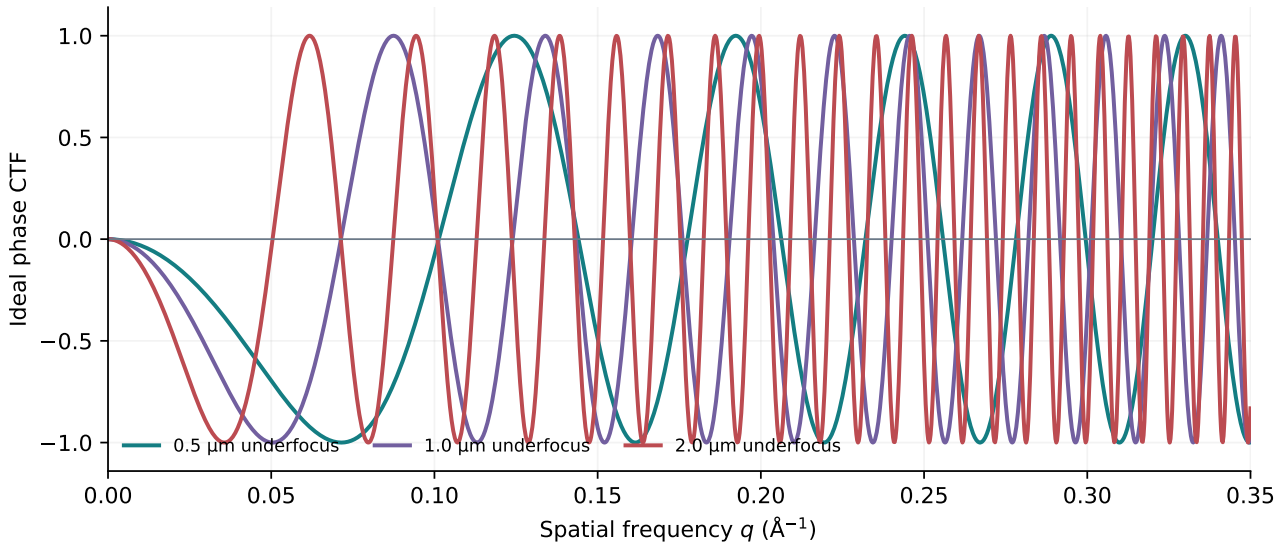
A MATHEMATICAL TEXTBOOK CHAPTER

From Electron Scattering to the Cryo-EM Contrast Transfer Function

Focus, defocus, wave optics, phase contrast, Thon rings, and CTF-aware single-particle reconstruction



What this chapter provides

An equation-by-equation account beginning with camera focusing and electron waves, proceeding through scattering and the weak-phase approximation, and ending with the CTF model and its use in CryoSPARC. Includes original diagrams, eight worked examples, conceptual checks, and exercises with solutions.

October 2026

Study guide / structural biology methods

Reading roadmap

Key idea

Central claim. The specimen scatters the electron wave regardless of focus. The objective lens imposes a frequency-dependent phase transfer. Deliberate underfocus turns the specimen's otherwise weak phase signal into detectable intensity modulation. The CTF describes how strongly each spatial frequency contributes to the measured image. The microscope creates the CTF; software estimates and compensates for it.

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Learning outcomes

After this chapter you should be able to (i) distinguish a focal point from an image plane; (ii) explain why changing objective-lens current changes the focus; (iii) derive scattering angles and the exit-wave phase; (iv) derive the weak phase object approximation and the pure-phase in-focus intensity result; (v) derive the Fresnel defocus phase and phase CTF; (vi) interpret CTF zeros and Thon rings; and (vii) describe how CryoSPARC estimates and uses per-particle CTFs.

Notation and conventions

Symbol	Meaning	Symbol	Meaning
λ	Electron wavelength	Δf	Defocus (positive for underfocus here)
$\mathbf{q}$	2D spatial frequency, Å^{-1}	$q = \lvert\mathbf{q}\rvert$	Spatial frequency magnitude
$V(\mathbf{r})$	Specimen electrostatic potential	$V_p(x, y)$	Projected potential
φ	Specimen phase shift	χ	Lens/defocus phase aberration
ψ	Complex electron wavefunction	C_s	Spherical-aberration coefficient
σ	Interaction constant	a	Amplitude-contrast coefficient
I	Recorded intensity	R_i	Particle orientation

Important qualification

We use the Fourier-transform kernel $e^{-2\pi i \mathbf{q}\cdot\mathbf{r}_\perp}$ and the standard positive-underfocus CryoSPARC phase convention $\chi(q) = -\pi\lambda\Delta f q^2 + (\pi/2)C_s\lambda^3 q^4$. The sign of image contrast also depends on how the ideal specimen signal is defined. Some texts use the negative of this CTF; that is a convention rather than a change of physics.

1 Focus: begin with an ordinary lens

1.1 Three distinct locations: object, lens, image

Think of a dot on a sheet of paper, a converging glass lens, and a camera sensor. Light leaves the dot in many directions. The lens changes the directions of the light rays. When the sensor coincides with the plane at which those rays converge, it records a sharp representation of the dot. If the sensor is positioned before or after that convergence plane, a point spreads over a finite blur region. That mismatch is *defocus*.

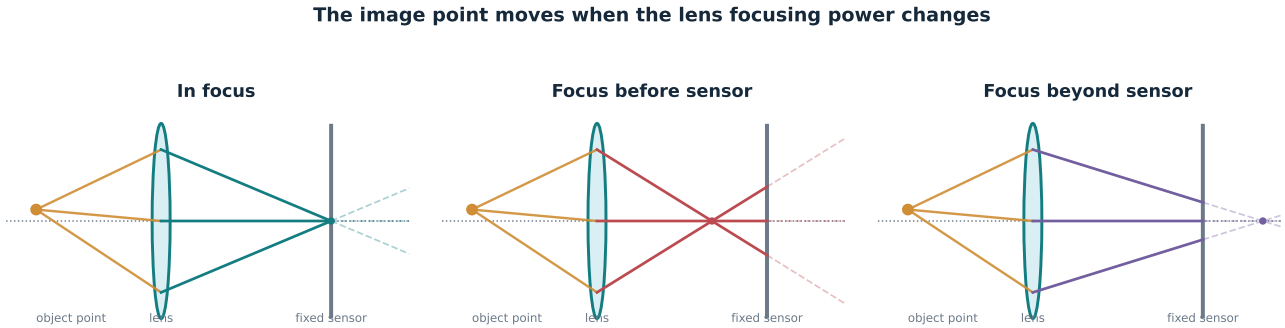


Figure 1: Original ray-optics illustration. The three panels have the same object and fixed sensor, but a different lens focusing strength. These are idealized ray diagrams used to define defocus; they are not a literal portrayal of a TEM's specimen-side underfocus convention.

An ordinary thin lens satisfies

$$\frac{1}{f_L} = \frac{1}{s} + \frac{1}{s'}, \qquad s' = \frac{s f_L}{s - f_L}, \tag{1}$$

where f_L is the focal length, s the object-to-lens distance, and s' the image distance. If the sensor is placed at s', it records the focused image. Changing f_L while holding the object and sensor fixed shifts the best image plane.

Worked example 1: move the image by changing the lens

Take an object $s = 30$ cm from a thin lens. With $f_L = 10$ cm, equation (1) gives

$$s' = \frac{30 \times 10}{30 - 10} = 15 \text{ cm}.$$

Place a sensor at 15 cm: the image is focused. Change the lens to $f_L = 12$ cm while keeping the sensor fixed:

$$s' = \frac{30 \times 12}{30 - 12} = 20 \text{ cm}.$$

The ideal image now forms 5 cm beyond the sensor. Thus the sensor records a defocused image. **Nothing about the object changed;** only the image-forming optics changed.

1.2 The focal point is not the same as an image point

A converging lens's *focal point* is defined for incoming rays that are parallel to the optical axis. A finite-distance object point, however, is generally imaged at a distance $s' \neq f_L$. Confusing these two points is a common source of difficulty.

There are also two distinct uses of focusing in an electron microscope: the condenser system controls illumination

of the specimen, while the objective lens controls how the transmitted wave forms the specimen image. In single-particle cryo-EM, objective *defocus* does not mean bringing the incoming beam to a small spot on the protein.

2 Electron lenses and the TEM image plane

2.1 How an electromagnetic lens focuses electrons

A transmission electron microscope (TEM) uses magnetic fields rather than glass to focus electron waves. An electron of charge $-e$ and momentum $\mathbf{p}$ experiences the Lorentz force

$$\mathbf{F} = -e(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \tag{2}$$

The curved trajectories in a shaped axial magnetic field act approximately like those in a converging lens. In an idealized rotationally symmetric magnetic lens, the paraxial focal strength has the form

$$\frac{1}{f_L} \simeq \frac{e^2}{4p^2} \int B_z(z)^2 \, \mathrm{d}z. \tag{3}$$

This is a schematic paraxial relation: real objectives require detailed magnetic-lens and aberration models. It tells us why changing the lens coil current changes the focusing power electronically.

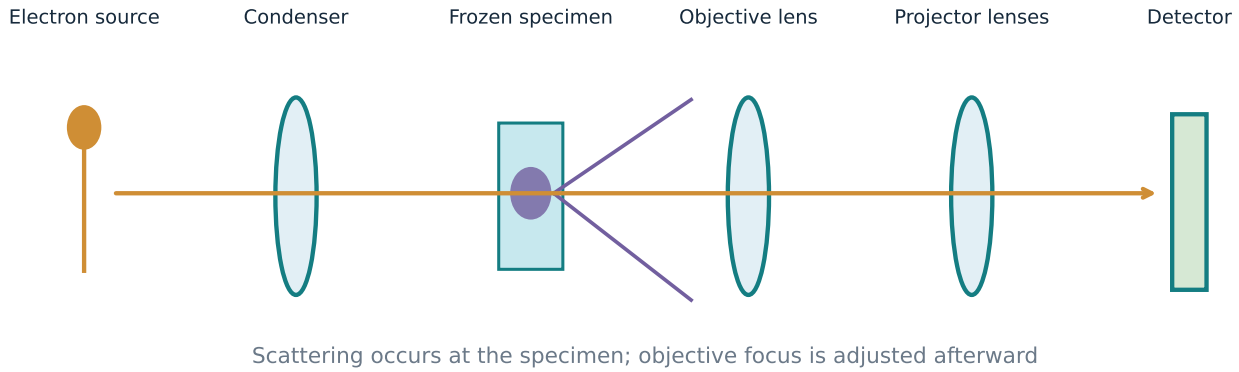


Figure 2: Simplified TEM signal path. The beam interacts with the frozen specimen *before* the objective lens establishes the image-formation condition. Additional lenses magnify and relay the image to the detector.

The objective lens has two conceptually important planes: its *back focal plane*, where different scattering angles are separated spatially, and its *image plane*, where the various transmitted and scattered components recombine into a real-space image. As a result, the lens both sorts angular components and causes them to interfere again.

2.2 The precise meaning of underfocus

In TEM convention, *underfocus* means slightly weaker objective-lens focusing than the setting that sharply images the actual specimen. Equivalently, the instrument is focused on a notional plane on the electron-source side of the specimen. For example, an intended underfocus of $\Delta f = 1.0\,\mu$m is an object-space displacement of

the focal setting by approximately that amount. It does *not* mean shifting the detector by a micron or moving the beam away from the protein.

Key idea

Focus is an **image-formation condition**. Scattering is a **sample interaction**. They occur in different parts of the TEM and must not be conflated.

3 Electron waves: amplitude, phase, wavelength

3.1 The electron as a coherent wave

A free electron wave with momentum $\mathbf{p} = \hbar\mathbf{k}$ may be written

$$\psi(\mathbf{r}, t) = A \exp\big[i(\mathbf{k}\cdot\mathbf{r} - \omega t)\big], \qquad |\mathbf{k}| = \frac{2\pi}{\lambda}. \tag{4}$$

The amplitude A and phase $\phi = \mathbf{k}\cdot\mathbf{r} - \omega t$ describe different properties. The detector measures an ensemble intensity proportional to $|\psi|^2$, not ϕ directly. Two waves can differ in phase but have the same measured intensity.

3.2 Relativistic wavelength at accelerating voltage U

The accelerated electron gains kinetic energy $K = eU$. Since $E_{\text{tot}}^2 = p^2c^2 + m_e^2c^4$ and $E_{\text{tot}} = m_ec^2 + K$,

$$(pc)^2 = K(K + 2m_ec^2), \tag{5}$$

$$\lambda = \frac{h}{p} = \frac{hc}{\sqrt{eU(eU + 2m_ec^2)}}. \tag{6}$$

The relativistic correction is essential at 200–300 kV; the nonrelativistic expression $h/\sqrt{2m_eeU}$ would be inaccurate.

Worked example 2: compute the wavelength at 300 kV

With $U = 300$ kV, $m_ec^2 = 511$ keV, and $hc = 12.398$ keV Å,

$$\lambda = \frac{12.398}{\sqrt{300(300 + 2\cdot 511)}}\ \text{Å} \approx 0.0197\ \text{Å}.$$

The wavelength is about 0.002 nm, far smaller than typical amino-acid or atomic spacings. That small wavelength makes very high scattering spatial frequencies physically accessible, although attainable reconstruction resolution is limited by many other factors.

3.3 A lens as a wavefront phase operator

Ray diagrams are useful but incomplete. In paraxial wave optics, an ideal thin lens transforms a transverse wave $\psi(x, y)$ by a quadratic phase factor:

$$L(x, y) = \exp\left[-\frac{i\pi}{\lambda f_L}(x^2 + y^2)\right]. \tag{7}$$

The lens therefore changes phase across the beam even if its transmitted amplitude is essentially unchanged. Subsequent propagation makes the components recombine at the appropriate image plane. This wavefront picture, rather than a literal collection of independent rays, is what we need to understand phase contrast.

4 Why the protein scatters the electron wave

4.1 The specimen is an electrostatic potential

Inside a protein, electrons encounter the electrostatic potential $V(x, y, z)$ produced by atomic nuclei and bound electrons. For an electron of charge $-e$ the potential energy is $-eV$. A basic nonrelativistic stationary equation illustrating the mechanism is

$$\left[-\frac{\hbar^2}{2m_e}\nabla^2 - eV(\mathbf{r})\right]\psi(\mathbf{r}) = E\psi(\mathbf{r}). \tag{8}$$

Relativistic corrections are necessary at cryo-EM voltages, but the implication is immediate: where V varies, a free plane wave is generally not an exact solution. The interaction produces outgoing wave components with different propagation directions.

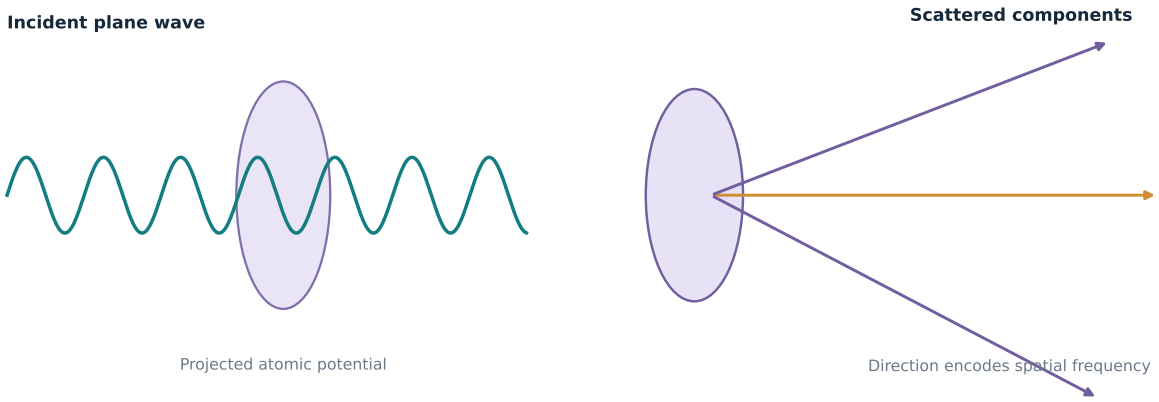


Figure 3: An incident electron wave interacts with a spatially varying specimen potential. The outgoing wave contains a strong forward component and weaker components propagating at other angles. The orange arrow is a reference direction, not a particular electron trajectory.

4.2 Scattering vector and scattering angle

Let $\mathbf{k}_i$ and $\mathbf{k}_f$ be the initial and final wavevectors. Define the momentum-transfer or scattering vector

$$\mathbf{Q} = \mathbf{k}_f - \mathbf{k}_i. \tag{9}$$

For elastic scattering, $|\mathbf{k}_i| = |\mathbf{k}_f| = 2\pi/\lambda$. Elementary geometry gives, for total deflection angle Θ,

$$|\mathbf{Q}| = \frac{4\pi}{\lambda}\sin\frac{\Theta}{2} \simeq \frac{2\pi}{\lambda}\Theta. \tag{10}$$

If a specimen feature has projected spatial frequency $q = 1/d$, the small-angle relationship is $\Theta \simeq \lambda q = \lambda/d$. Smaller feature sizes produce larger scattering angles.

4.3 Born scattering and the Fourier transform

In the first Born approximation, the coherent elastic scattering amplitude is proportional to the spatial Fourier transform of the specimen interaction potential:

$$\boxed{f(\mathbf{Q}) \propto \int_{\mathbb{R}^3} V(\mathbf{r}) e^{-i\mathbf{Q}\cdot\mathbf{r}}\, \mathrm{d}^3\mathbf{r}.} \tag{11}$$

Thus spatially varying molecular structure determines the distribution of scattered directions. This is a useful bridge from intuitive "scattering by atoms" to the Fourier-space image formation used in cryo-EM. *The first Born approximation is not exact*; dynamical or multiple scattering requires more complicated treatments.

Key idea

At focus and at defocus, the same specimen still produces scattering. Changing the objective focus changes the **interference and contrast after scattering**, not whether the specimen can scatter at all.

5 From the sample potential to the exit wave

5.1 Projecting the potential through a thin specimen

The basic single-particle model treats the sample as sufficiently thin that the electron's path is approximately straight and multiple scattering can be neglected. Define the projected electrostatic potential

$$V_p(x, y) = \int V(x, y, z)\, \mathrm{d}z. \tag{12}$$

A phase-object exit wave is then approximately

$$\psi_{\text{exit}}(x, y) = \psi_{\text{in}}(x, y) \exp[i\sigma V_p(x, y)], \tag{13}$$

where the relativistic electron interaction constant is $\sigma = e/(\hbar v)$ and v is the electron speed. With a unit incident plane wave, let $\varphi(x, y) = \sigma V_p(x, y)$ so that $\psi_{\text{exit}} = e^{i\varphi}$.

Worked example 3: potential to electron phase shift

At 300 kV, $\sigma \simeq 6.53\ \text{rad}/(\text{V}\,\mu\text{m})$. Suppose the projected potential along a ray is $V_p = 0.010\ \text{V}\,\mu\text{m}$. Then

$$\varphi = \sigma V_p = (6.53)(0.010) = 0.0653\ \text{rad} = 3.74^\circ.$$

The wave $\exp(i\varphi)$ has the *same modulus* as the incoming wave but a small phase difference. The small value makes the weak-phase approximation reasonable for this illustrative ray, though real specimens are spatially heterogeneous and may require more sophisticated models.

5.2 Derive the weak phase object approximation

When $|\varphi| \ll 1$, expand the exponential:

$$\psi_{\text{exit}} = e^{i\varphi} = 1 + i\varphi - \tfrac{1}{2}\varphi^2 - \tfrac{i}{6}\varphi^3 + \cdots, \tag{14}$$

$$\boxed{\psi_{\text{exit}} \simeq 1 + i\varphi.} \tag{15}$$

The first term is the unscattered reference component; the second is the small first-order scattered perturbation. Because $i = e^{i\pi/2}$, this perturbation is in quadrature (90 degrees in phase) with the reference, when φ is real.

Using $\widetilde{g}(\mathbf{q}) = \int g(\mathbf{r}_\perp) e^{-2\pi i \mathbf{q}\cdot\mathbf{r}_\perp}\, \mathrm{d}^2\mathbf{r}_\perp$,

$$\widetilde{\psi}_{\text{exit}}(\mathbf{q}) \simeq \delta_D(\mathbf{q}) + i\widetilde{\varphi}(\mathbf{q}). \tag{16}$$

The delta function represents the central reference beam, while the nonzero-frequency term is the scattered component. The phase-object and scattering descriptions are therefore the same mathematics viewed in real versus Fourier space.

5.3 Why the exact pure phase object is invisible in focus

At an ideal in-focus exit-wave image plane, a perfect pure-phase object gives

$$I(x, y) = |e^{i\varphi(x,y)}|^2 = 1. \tag{17}$$

The spatial phase pattern can be complicated, but the intensity is uniform. The detector records no phase contrast from this idealized object. If one instead substitutes $1 + i\varphi$ and obtains $1 + \varphi^2$, the φ^2 term is *beyond the first-order accuracy* of the approximation: the exact phase-only result remains unity.

Important qualification

This does not say that real in-focus cryo-EM images are completely uniform. Small amplitude contrast, spherical aberration, inelastic effects, phase plates if present, and other imperfections can still create measurable contrast. "Zero in-focus contrast" refers to the ideal pure-phase, aberration-free first-order result.

6 Defocus as a frequency-dependent phase shift

6.1 Derive the Fresnel propagation factor

Consider a plane-wave component with transverse spatial frequency $\mathbf{q} = (q_x, q_y)$. Its transverse wavevector is $\mathbf{k}_\perp = 2\pi\mathbf{q}$. Elastic dispersion requires

$$k_z^2 + |\mathbf{k}_\perp|^2 = k_0^2, \qquad k_0 = \frac{2\pi}{\lambda}. \tag{18}$$

For small propagation angles,

$$k_z = \sqrt{k_0^2 - (2\pi q)^2} \tag{19}$$

$$= k_0\sqrt{1 - \lambda^2 q^2} \tag{20}$$

$$\simeq k_0 - \frac{k_0\lambda^2 q^2}{2} = k_0 - \pi\lambda q^2. \tag{21}$$

Over a distance z, the component gains a factor $e^{ik_z z}$. Dropping the common carrier phase $e^{ik_0 z}$ yields

$$\boxed{H_z(q) = e^{-i\pi\lambda z q^2}.} \tag{22}$$

This is the paraxial *Fresnel propagator*. The same quadratic phase structure describes the effective departure of an image from exact focus. For positive-underfocus Δf,

$$\boxed{\chi_{\mathrm{def}}(q) = -\pi\lambda\Delta f q^2.} \tag{23}$$

The direct lesson is that defocus does not add the same phase to all waves: the extra phase depends on q^2. A low spatial frequency and a high spatial frequency can therefore interfere differently with the central reference wave in the final image.

6.2 Defocus has a quadratic dependence on spatial frequency

Doubling defocus doubles the quadratic phase. Doubling spatial frequency multiplies it by four:

$$\chi_{\mathrm{def}}(2q) = 4\chi_{\mathrm{def}}(q).$$

A lens focused away from the specimen therefore changes different structural details by different amounts. This is the beginning of the oscillatory contrast transfer function.

7 Derive intensity contrast from the scattered wave

7.1 From exit wave to detected intensity

Start with $\psi_{\mathrm{exit}} \simeq 1 + i\varphi$. Let $H(\mathbf{q}) = e^{i\chi(\mathbf{q})}$ be the effective phase transfer, with $\chi(0) = 0$ and $\chi(-\mathbf{q}) = \chi(\mathbf{q})$ for the symmetric aberrations considered here. The image-plane wave is

$$\psi_{\mathrm{img}} \simeq 1 + i\mathcal{F}^{-1}\left[e^{i\chi(\mathbf{q})}\widetilde{\varphi}(\mathbf{q})\right]. \tag{24}$$

Let $g = \mathcal{F}^{-1}[e^{i\chi}\widetilde{\varphi}]$. Expanding the intensity and discarding second-order terms,

$$I = |1 + ig|^2 \simeq 1 + i(g - g^*) \tag{25}$$

$$= 1 - 2\,\mathrm{Im}\, g. \tag{26}$$

Because φ is real and χ is even, the imaginary part comes from $\sin\chi$:

$$\boxed{\widetilde{\Delta I}(\mathbf{q}) \simeq -2\sin\chi(\mathbf{q})\,\widetilde{\varphi}(\mathbf{q}), \quad \mathbf{q} \neq 0.} \tag{27}$$

If we define an ideal specimen contrast signal $\widetilde{S} = -2\widetilde{\varphi}$, the observable transfer becomes $\widetilde{\Delta I} = \sin\chi\,\widetilde{S}$. This choice matches the positive $\sin\chi$ CryoSPARC CTF convention.

7.2 What is physically changing in defocus?

The specimen's wave interaction is already complete. Defocus changes the *relative phase* of the scattered Fourier components as they recombine with the unscattered reference. Constructive and destructive interference now create measurable intensity fluctuations. The underlying scattering is not increased merely by turning the focus knob.

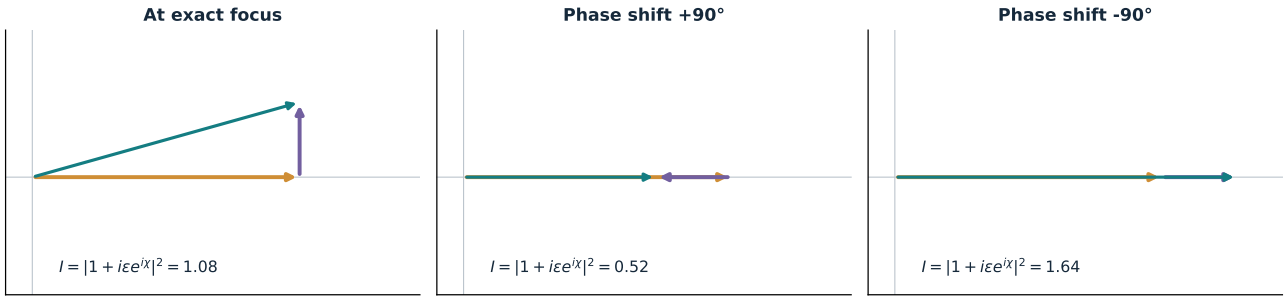


Figure 4: Original phasor comparison. The gold vector is the unscattered reference; purple is a small scattered addition; teal is the summed wave. At focus (left), the scattered perturbation is nearly perpendicular to the reference and produces only second-order intensity differences. Additional relative phase can increase or decrease the measured intensity. For illustrative $\epsilon = 0.28$.

7.3 Optional real-space check: transport of intensity

The paraxial envelope equation is $\partial_z \psi = (i/2k_0)\nabla_\perp^2 \psi$. Writing $\psi = \sqrt{I}\, e^{i\varphi}$ and separating real and imaginary parts gives the transport-of-intensity equation

$$\frac{\partial I}{\partial z} = -\frac{1}{k_0}\nabla_\perp \cdot (I \nabla_\perp \varphi). \tag{28}$$

For nearly uniform $I \simeq 1$ and small propagation z,

$$I_z \simeq 1 - \frac{\lambda z}{2\pi}\nabla_\perp^2 \varphi. \tag{29}$$

The intensity change is proportional to the *spatial curvature* of phase. A constant phase offset produces no first-order image contrast. This agrees with the small-z expansion of $\sin\chi$ from equation (27).

8 The contrast transfer function

8.1 Exact definition within the linear weak-object model

The CTF is a frequency-domain response for the image contrast:

$$\boxed{\widetilde{\Delta I}(\mathbf{q}) = \mathrm{CTF}(\mathbf{q})\,\widetilde{S}(\mathbf{q}) + \widetilde{N}(\mathbf{q}).} \tag{30}$$

The specimen supplies $\widetilde{S}$; the optics supplies the CTF; the detector and acquisition introduce noise $\widetilde{N}$. The CTF is *not* a digitally chosen post-processing filter. It describes the actual imaging physics that must be modeled later.

Equivalently, with a linear shift-invariant point-spread function h, $\Delta I = h * S + N$ and $\mathrm{CTF} = \widetilde{h}$. This equivalence is useful when moving between real-space blur and Fourier-space transfer.

8.2 Add spherical aberration

A real objective lens also has spherical aberration. Its phase contribution grows as q^4. With spherical-aberration coefficient C_s, the phase function is

$$\boxed{\chi(q) = -\pi\lambda\Delta f\, q^2 + \frac{\pi}{2}C_s\lambda^3 q^4 + \eta,} \tag{31}$$

where η is an optional relative phase-plate shift. The ideal phase CTF is

$$\boxed{\mathrm{CTF}_{\mathrm{phase}}(q) = \sin \chi(q).} \tag{32}$$

In physical terms: the q^2 term is caused by defocus, the q^4 term by spherical aberration, and η by a phase plate if present. These features appear explicitly in the CryoSPARC contrast guide [1].

8.3 Amplitude contrast and envelopes

Real specimens have some amplitude contrast. One common extension of the model is

$$\mathrm{CTF}_{\mathrm{full}}(q) = E(q)\left[\sqrt{1-a^2}\sin \chi(q) - a\cos \chi(q)\right], \tag{33}$$

where $0 \le a \le 1$ is an amplitude-contrast fraction and $E(q)$ is a damping envelope. Other packages may use different overall signs or phase conventions. The envelope represents, for example, temporal/spatial coherence and other frequency-dependent losses, and is often smaller at high q.

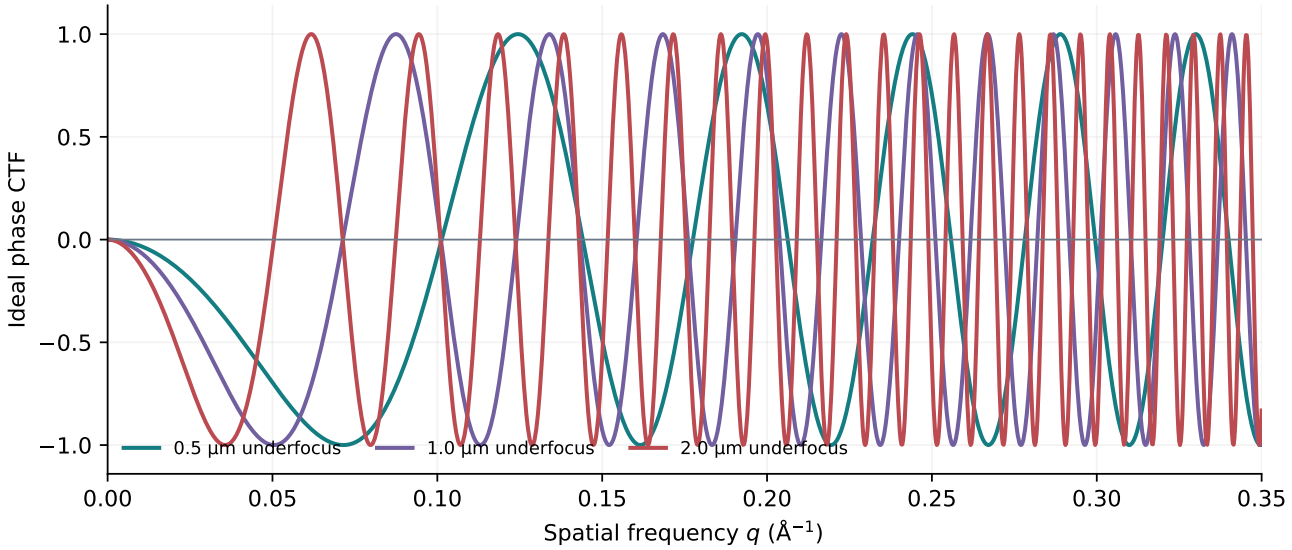


Figure 5: Original ideal CTF curves at 300 kV, $C_s = 2.7$ mm, with 0.5, 1.0, and 2.0 μm underfocus. The curve oscillates, reverses sign, and has zeros. The figure does not include amplitude contrast, envelope damping, or a phase plate.

8.4 Interpreting zero, positive, and negative CTF values

At $\mathrm{CTF}(q) \approx +1$, the specimen frequency is strongly transferred with one sign. At -1, it is transferred with reversed sign. Near zero, the first-order signal is suppressed regardless of whether the specimen actually contains that frequency. This last point is crucial: *CTF zeros are losses of observable information, not evidence that the protein lacks the corresponding structures.*

Worked example 4: 20-Å and 10-Å features at 1 μm underfocus

At 300 kV, take $\lambda = 0.0197$ Å and $\Delta f = 1\,\mu\mathrm{m} = 10^4$ Å. Ignore C_s so that $\chi = -\pi\lambda\Delta f q^2$.

A 20-Å feature has $q = 1/20 = 0.05$ Å^{-1}:

$$\chi_{20} = -\pi(0.0197)(10^4)(0.05)^2 = -1.547, \qquad \mathrm{CTF} \simeq -1.000.$$

A 10-Å feature has $q = 0.10$ Å^{-1}:

$$\chi_{10} = -\pi(0.0197)(10^4)(0.10)^2 = -6.189, \qquad \mathrm{CTF} \simeq +0.094.$$

The protein scatters at *both* frequencies; the chosen defocus strongly transmits one and almost suppresses the other. The particular values change when C_s and amplitude contrast are included.

8.5 Locations of CTF zeros

Ignoring C_s and phase-plate shift, $\sin(-\pi\lambda\Delta f q^2) = 0$ at nonzero frequencies

$$q_n = \sqrt{\frac{n}{\lambda\Delta f}}, \qquad d_n = \frac{1}{q_n} = \sqrt{\frac{\lambda\Delta f}{n}}, \qquad n = 1, 2, \ldots \tag{34}$$

These zeros shift as defocus changes.

Worked example 5: location of the first CTF zero

With $\lambda = 0.0197$ Å and $\Delta f = 10^4$ Å, the first nonzero crossing is

$$q_1 = \sqrt{\frac{1}{0.0197 \cdot 10^4}} = 0.07125 \text{ Å}^{-1}, \quad d_1 = 14.04 \text{ Å}.$$

In a spectrum from this simplified system, spatial information near 14 Å will have little first-order phase contrast. Other defocus values move that missing band.

9 2D CTF, astigmatism, and Thon rings

9.1 Why a single radial curve is sometimes insufficient

Real micrographs are two-dimensional, so frequencies are vectors $\mathbf{q} = (q_x, q_y)$. A perfectly rotationally symmetric objective has the same defocus for every Fourier-space direction. Real objectives often exhibit *astigmatism*, meaning the focusing strength depends on angle.

Let Δf_U and Δf_V be principal-axis defoci, θ_{ast} the principal orientation, and $\theta = \mathrm{atan2}(q_y, q_x)$. A conventional angular parameterization is

$$\Delta f(\theta) = \frac{\Delta f_U + \Delta f_V}{2} + \frac{\Delta f_U - \Delta f_V}{2}\cos\big(2(\theta - \theta_{\mathrm{ast}})\big). \tag{35}$$

Replace Δf with $\Delta f(\theta)$ in equation (31) to obtain $\mathrm{CTF}(q, \theta)$.

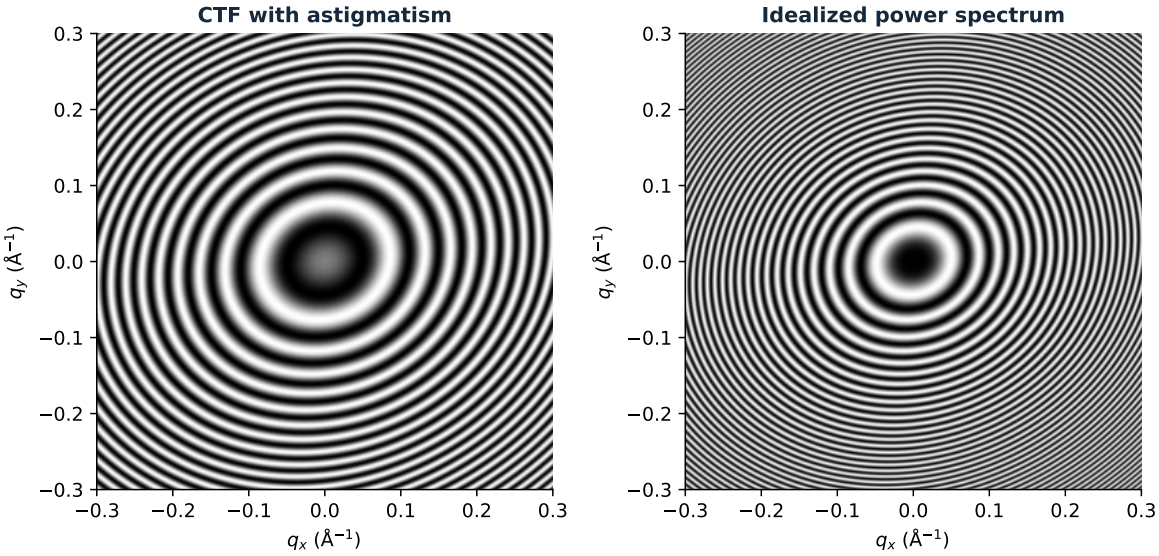


Figure 6: Original two-dimensional simulation with 300 kV, $C_s = 2.7$ mm, principal defoci 1.0 and 1.35 μm, and nonzero astigmatism angle. The left is signed CTF; the right is its squared magnitude, which resembles the oscillatory modulation measured in a micrograph power spectrum.

9.2 Why Thon rings appear in image power spectra

For recorded Fourier image $Y(\mathbf{q}) = C(\mathbf{q})S(\mathbf{q}) + N(\mathbf{q})$, the power spectrum is $|Y|^2$. If signal and noise are uncorrelated,

$$\boxed{\mathbb{E}|Y(\mathbf{q})|^2 \simeq |C(\mathbf{q})|^2 P_S(\mathbf{q}) + P_N(\mathbf{q}).} \tag{36}$$

The $|C|^2$ oscillation creates alternating bright and dim rings called *Thon rings*. Astigmatism makes the rings elliptical rather than circular. Because power depends on C^2, the raw ring contrast does not directly reveal whether C is positive or negative; the physical CTF model provides the sign.

Worked example 6: astigmatism gives directional CTF differences

Suppose $\Delta f_U = 1.00\,\mu$m and $\Delta f_V = 1.40\,\mu$m. Along the first principal axis $\theta = \theta_{\text{ast}}$, equation (35) gives $\Delta f = 1.00\,\mu$m; along the perpendicular axis it gives $1.40\,\mu$m. At $q = 0.10$ Å^{-1} and neglecting C_s, the phase is $\chi = -\pi\lambda\Delta f q^2$. It is approximately -6.19 rad in one direction and -8.66 rad in the other. Therefore the transfer oscillations have *different phases in different directions*. Fitting only a rotationally averaged 1D curve would throw away the astigmatism information.

10 How experimental CTF estimation works

10.1 What the microscope operator changes

The microscope operator or automated acquisition software sets target defocus by adjusting the objective-lens current. In conventional single-particle cryo-EM, a range such as 0.5–2.0 μm underfocus may be used for an illustrative example, though optimal values vary substantially with specimen and imaging conditions. The beam still illuminates the specimen, and the detector records intensity movie frames. Focus choice is *not* a digital operation on already collected data.

10.2 What CryoSPARC estimates afterward

After motion correction, CryoSPARC's *Patch CTF Estimation* fits CTF-like oscillations in each micrograph's Fourier power spectrum, estimating defocus, astigmatism, and a smoothly varying defocus landscape. CryoSPARC's documentation describes local patches, 2D fitting, CTF envelopes and optimization [3, 4].

One illustrative estimation model is

$$P_{\mathrm{model}}(\mathbf{q};\boldsymbol{\theta}) = B(\mathbf{q}) + A(\mathbf{q})E(\mathbf{q})^2|C(\mathbf{q};\boldsymbol{\theta})|^2, \tag{37}$$

where B is background power, A is a specimen-spectrum factor, and $\boldsymbol{\theta} = (\Delta f_U, \Delta f_V, \theta_{\mathrm{ast}}, \ldots)$ collects optical parameters. A conceptual fit minimizes mismatch between P_{model} and observed power; *this is explanatory, not a transcription of CryoSPARC's exact internal optimization loss.*

10.3 Why particle defocus is not always identical across a micrograph

The support/ice film can be tilted or curved; particles can occupy different heights. Hence the effective defocus can depend on image position:

$$\Delta f = \Delta f(x, y), \qquad \Delta f_i = \Delta f(x_i, y_i). \tag{38}$$

Patch CTF provides a spatial model. In later high-resolution processing, *Local CTF Refinement* can refine per-particle defocus against a reference volume, while global refinement can address higher-order aberrations [5].

10.4 How to interpret CryoSPARC diagnostics

In the Patch CTF output, compare the observed power-spectrum oscillations to the fitted oscillatory model. The fit-resolution indicator is an empirical measure of visible CTF signal rather than a hard final reconstruction-resolution limit. CryoSPARC notes that a particular fit correlation threshold (often 0.3 in the diagnostic plot) is used for this readout [3]. A bad fit can arise from poor ice, strong motion, insufficient signal, large astigmatism, or other image problems; neither a single fitted defocus number nor a single fit-resolution value is sufficient to assess all data quality.

Key idea

> Experimental sequence: **set objective underfocus** → **record movies** → **motion-correct micrographs** → **fit Thon-ring/CTF patterns** → **attach CTF parameters to particles** → **CTF-aware refinement**.

11 Why the CTF must be corrected computationally

11.1 A CTF is not an innocuous blur

In Fourier space, let $Y(q) = C(q)S(q) + N(q)$. When $C > 0$, image contrast has one sign; for $C < 0$, the contrast is reversed; when $C \approx 0$, the signal is very weak. Ignoring C assumes every spatial frequency is equally and correctly transferred, which is false.

Naive inversion would be

$$\widehat{S}_{\mathrm{naive}}(q) = \frac{Y(q)}{C(q)} = S(q) + \frac{N(q)}{C(q)}. \tag{39}$$

If $C \to 0$, noise is amplified without bound. If $C = 0$ exactly, the observation contains no first-order information about $S(q)$ and inversion is impossible from that exposure alone.

11.2 Why varying defocus helps

Different defocus values shift the CTF zeros. Thus one exposure may carry a frequency which another suppresses. This is useful for combining many particles and datasets, although different particle orientations and the three-dimensional Fourier sampling geometry must also be accounted for.

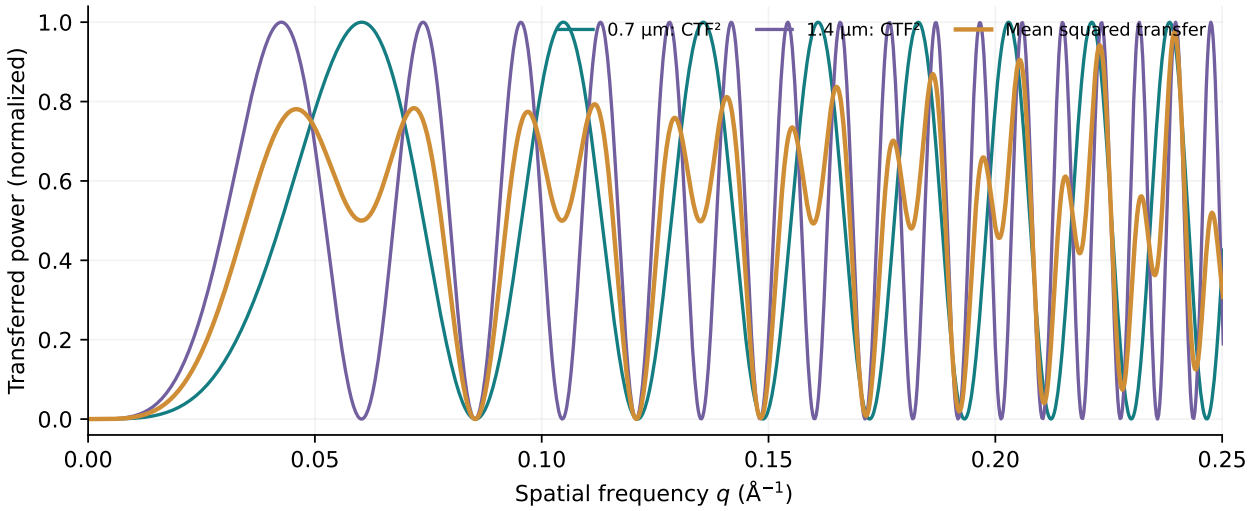


Figure 7: Original illustration of complementary CTF transfer at 0.7 and 1.4 μm underfocus. The curves show squared transfer for each condition and the average of the two, not actual experimental signal-to-noise ratios.

11.3 Least-squares and Wiener-like combination

For multiple observations of the same Fourier component, $Y_i = C_i S + N_i$ with independent zero-mean Gaussian noise of variance σ_i^2, regularized weighted least squares yields

$$\boxed{\widehat{S} = \frac{\sum_i C_i^* Y_i / \sigma_i^2}{\sum_i |C_i|^2 / \sigma_i^2 + \tau^{-2}},} \tag{40}$$

where τ^2 is an assumed prior variance on S. The numerator uses the sign and phase of the CTF; the denominator downweights poorly transferred frequencies and regularizes nearly unobservable ones. Actual cryo-EM refinement solves a more complicated 3D estimation problem, but this scalar example captures the essential logic.

Worked example 7: two defocus settings recover a missing frequency

Suppose the true Fourier coefficient is $S = 2$, and the first image has $C_1 = 0$, giving $Y_1 = 0.10$ from noise alone. The second image has $C_2 = 0.8$ and observed $Y_2 = 1.70$. Ignoring a prior and assuming equal noise variances,

$$\widehat{S} = \frac{C_1 Y_1 + C_2 Y_2}{C_1^2 + C_2^2} = \frac{0 + 0.8(1.70)}{0 + 0.64} = 2.125.$$

The first observation contributes no information at that frequency, but the second recovers it with some noise. Simply dividing the first observation by its zero CTF would be undefined.

12 The single-particle reconstruction forward model

12.1 A particle image is a projection, not a slice of the object in real space

Let $V(\mathbf{r})$ denote a 3D specimen potential, and R_i the orientation of particle i. A projection is

$$(P_{R_i}V)(x,y) = \int V\left(R_i^{-1}(x,y,z)\right)\,\mathrm{d}z. \tag{41}$$

By the Fourier slice theorem, its 2D Fourier transform samples a central slice of the 3D Fourier transform $\widetilde{V}$.

With in-plane translation $\mathbf{t}_i$, CTF $C_i(\mathbf{q})$, and noise $N_i(\mathbf{q})$, one useful model is

$$\boxed{Y_i(\mathbf{q}) = C_i(\mathbf{q})\,\widetilde{V}\left(R_i^{\mathsf{T}}(q_x,q_y,0)^{\mathsf{T}}\right)e^{-2\pi i\mathbf{q}\cdot\mathbf{t}_i} + N_i(\mathbf{q}).} \tag{42}$$

It is understood that overall intensity scaling and sign conventions have been absorbed in the definition of V and C_i. This model is the bridge between electron optics and the inverse problem solved in modern SPA refinement.

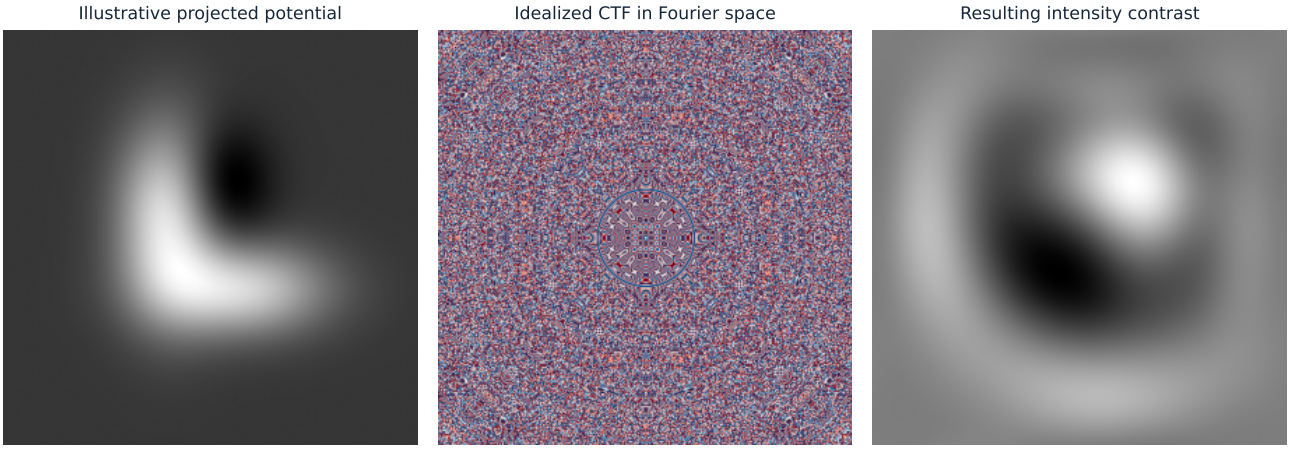


Figure 8: Synthetic forward-model illustration: a projected specimen signal, a signed 2D CTF, and the resulting image contrast. The spatial-frequency pattern has been shifted for display; it is illustrative and not a realistic noisy micrograph.

12.2 Why CTF-aware reconstruction is preferable to perfect-looking 2D correction

A reconstruction algorithm can incorporate C_i directly into each particle's forward prediction and compare that prediction to the recorded image. It is not necessary to first divide every raw particle by its CTF. Working within the forward model avoids uncontrolled noise amplification at the CTF zeros and allows observations with complementary orientations and defocus settings to contribute where informative.

12.3 Connection to your structural-analysis work

If a PP2A A-subunit DNA-spring sample exhibits small shape changes, a mismodeled CTF can bias the apparent appearance of densities or impede particle alignment. In any heterogeneous refinement or neural reconstruction, use the particle's actual $C_i(\mathbf{q})$ in the image likelihood or forward model. Otherwise, the model might confuse image-formation differences between particles with true protein conformational heterogeneity. CTF correction is necessary but not sufficient: orientation bias, motion, conformational variability, and sample heterogeneity must still be addressed separately.

13 Worked integration problem: from focal setting to map

Worked example 8: an end-to-end cryo-EM imaging decision

Scenario. You image a flexible protein at 300 kV. Your microscope has $C_s = 2.7$ mm. You acquire particles at nominal underfocus 0.7 and 1.4 μm. You want to determine whether a small domain is consistently visible.

Step 1: Convert quantities. The electron wavelength is about 0.0197 Å. Write $C_s = 2.7 \times 10^7$ Å, and $\Delta f = 7000$ or 14000 Å.

Step 2: Choose a detail scale. A 10-Å feature has $q = 0.10$ Å^{-1}.

Step 3: Compute a phase function.

$$\chi(q) = -\pi(0.0197)\Delta f(0.10)^2 + \frac{\pi}{2}(2.7 \times 10^7)(0.0197)^3(0.10)^4.$$

The C_s contribution at this frequency is approximately $+0.032$ rad. The defocus contributions are about -4.33 rad at 0.7 μm and -8.66 rad at 1.4 μm.

Step 4: Predict CTF transfer. Their sine values are approximately $+0.92$ and -0.71, respectively. Thus the same structural feature can appear with a different sign and different magnitude in different micrographs.

Step 5: Process correctly. Motion-correct the movies, fit CTFs, attach estimates to particle metadata, and reconstruct using the forward model $Y_i = C_i P_{R_i} V + N_i$. Do not regard a contrast reversal across micrographs as proof that the protein physically changed.

Step 6: Validate. Examine CTF diagnostics, orientations, refinement stability and independent half-map consistency before interpreting a domain as conformationally altered.

14 Common misconceptions, corrected

Misconception	Correct interpretation
"Defocus causes scattering."	Scattering is produced by the specimen electrostatic potential; defocus changes how scattered and unscattered components interfere.
"At focus, no electrons scatter."	Scattering still occurs, but ideal weak pure-phase contrast is zero to first order.
"Focus means squeezing the beam onto the protein."	Condenser illumination and objective image focus are different controls.
"CTF zero means the protein lacks that structural detail."	The frequency is suppressed by the optics, not necessarily absent in the specimen.
"CTF is a filter the software applies to raw data."	The microscope imposes transfer during exposure; software estimates and compensates for it.
"Divide by CTF to undo blur."	Near zeros, direct division catastrophically amplifies noise.
"Thon rings directly reveal signed CTF."	The power spectrum is modulated by approximately $\lvert \mathrm{CTF} \rvert^2$; sign comes from the physical model.
"A high CTF-fit resolution guarantees a high-resolution map."	A good fit helps, but motion, orientation distribution, heterogeneity, and many other factors limit map quality.

15 Self-test exercises

Q1. For a thin glass lens, let $s = 40$ cm and $f_L = 8$ cm. Where does the sharp image form? If the sensor remains there and f_L changes to 10 cm, how far away does the new best image form?

Q2. In $\psi_{\text{exit}} = e^{i\varphi}$, why does intensity remain exactly 1 for a pure phase object? Why is $|1 + i\varphi|^2 = 1 + \varphi^2$ not a contradiction?

Q3. Show using $k_z = \sqrt{k_0^2 - (2\pi q)^2}$ that the paraxial free-space phase relative to the central beam is $-\pi\lambda z q^2$.

Q4. In the $C_s = 0$ model, by what factor does the first CTF zero frequency change when underfocus increases from 0.5 to 2.0 μm?

Q5. Why can a rotationally averaged power spectrum conceal astigmatism? Which additional fitted parameters are needed?

Q6. Suppose the image signal is $Y = CS + N$, $C = 0.01$, and noise standard deviation 0.1. What happens to the noise standard deviation if you perform naive CTF division?

Q7. What two physical differences exist between a CTF-free ideal projection and an actual particle micrograph, even if they have the same orientation and translation?

Q8. Can a map be recovered at a Fourier frequency where all available CTFs are exactly zero? Explain the role of priors versus experimental evidence.

16 Selected solutions

A1. $s' = sf_L/(s - f_L)$ gives 10 cm initially and $400/30 = 13.33$ cm afterward. The best image plane moves about 3.33 cm beyond the fixed sensor.

A2. $|e^{i\varphi}| = 1$ exactly. Truncating the wave to first order and then retaining a second-order intensity term is an inconsistent approximation; the exact exponential contains terms that cancel it.

A3. Write $k_z = k_0\sqrt{1 - (2\pi q/k_0)^2}$ and use $\sqrt{1-x} \simeq 1 - x/2$. With $k_0 = 2\pi/\lambda$, $k_z - k_0 = -\pi\lambda q^2$. Multiply by z.

A4. $q_1 \propto (\Delta f)^{-1/2}$, so increasing underfocus by a factor of four reduces the first zero frequency by a factor of two.

A5. Astigmatism changes Δf with Fourier-space angle. Radial averaging discards angle. Fit Δf_U, Δf_V, and θ_{ast}.

A6. Direct division scales noise as $\sigma_N/|C| = 0.1/0.01 = 10$, a hundredfold amplification.

A7. The actual micrograph is CTF-modulated and noisy. It may additionally reflect motion and other aberrations beyond the simple model.

A8. No direct first-order evidence remains at that frequency. Priors or regularizers may supply a plausible estimate, but cannot create measurement information that all CTFs removed.

17 References and further reading

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